

- b) Find the Fourier transform of $f(x) = e^{-x^2}$. 7

OR

Find Fourier sine and cosine transform of $f(x) = e^{-ax}$ ($a > 0$).

5. a) Find the solution of one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ 8

where $u(x,t)$ is the deflection of the string, with boundary condition $u(0,t) = 0$ and $u(L,t) = 0$ for all t and initial condition

$$u(x,0) = f(x)$$

$$\frac{\partial u}{\partial t} \text{ (at } t = 0) = g(x).$$

- b) Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0 assuming that the initial temperature is 7

$$f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$$

6. a) Write one-dimensional heat equation and solve it completely. 7

OR

Define potential function and find its solution by spherical membrane.

- b) Maximize $z = 150x_1 + 300x_2$ subject to $2x_1 + x_2 \leq 16$, $x_1 + x_2 \leq 8$, $x_2 \leq 3.5$, $x_1 \geq 0$ and $x_2 \geq 0$. 8

7. Attempt the following questions:

- a) Find $Z(1)$ 2
- b) Show that $F_s\{a f(x) + b g(x)\} = a F_s\{f(x)\} + b F_s\{g(x)\}$ where F_s stands for the Fourier sine transform. 2
- c) Solve the partial differential equation $u_{xx} - u_{yy} = 0$, by using separation of variables methods. 2
- d) Find $Z^{-1}[z/(z-3)]$ 2
- e) What do you mean by linear programming? 2

Level: Bachelor
 Programme: BE
 Course: Engineering Mathematics IV

Semester – Fall

Year : 2012
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) State Cauchy Riemann equations. Using these equations, show that the function $f(z) = z^3$ is everywhere analytic. 8
- b) Write the wave equation for vibrating circular membrane together with its initial and boundary conditions and solve it. 7

OR

State and prove Cauchy's integral theorem.

2. a) Obtain the Fourier integral formula from the Fourier series assuming the required conditions. 8

OR

Show that:
$$\int_0^\infty \left(\frac{1 - \cos \pi w}{w} \right) \sin wx dw = \begin{cases} \pi/2 & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

- b) Find the Fourier transform of $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ 7
3. a) State and prove first shifting theorem of Z – transform and hence find $Z(\cos at)$ and $Z(\sin at)$. 8
- b) Solve the difference equation by using Z – transform $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$ where $y_0 = 0, y_1 = 1$ 7
4. a) Derive the one dimensional wave equation
$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$
 8
- b) Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial 7

temperature is

$$f(x) = \begin{cases} x & \text{if } 0 < x < L/2 \\ L - x & \text{if } \frac{L}{2} < x < L. \end{cases}$$

5. a) Solve the following linear programming problem by using simplex method.
Minimize $z = 4x_1 + 3x_2$ subjected to $2x_1 + 3x_2 \geq 1$, $3x_1 + x_2 \geq 4$, $x_1 \geq 0$, $x_2 \geq 0$.
- b) Using the method of separation of variable find the solution $u(x, y)$ of the partial differential equation $xu_{xy} + 2yu = 0$.
6. a) Find the tangent vector to the curve $\vec{r}(t) = 2\cos t \vec{i} + \sin t \vec{j}$ at $(\sqrt{2}, \sqrt{2}, 0)$. Also find the tangent at the given point.
- b) Define Z- transform of a function $f(t)$ and by using the definition find the Z- transform of (i) $(-1)^n$ (ii) $\frac{1}{n!}$
- c) Write a short note on Linear Programming.
7. Attempt all the questions:
- a) Express $f(z) = \sin z$ in the form $u + iv$.
- b) Evaluate $\oint_c \frac{dz}{z}$ where c is the unit disk $|z| = 1$.
- c) Show that $\mathcal{F}_s\{af(x) + bg(x)\} = a\mathcal{F}_s\{f(x)\} + b\mathcal{F}_s\{g(x)\}$, where $\mathcal{F}_s$ stand for Fourier sine transform.
- d) Find the parametric representation of the surface $x^2 + 4y^2 = 9$, $z = 3$.

Level: Bachelor
 Programme: BE
 Course: Engineering Mathematics IV

Semester: Spring

Year : 2012
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

4.

- a) Define analytic function $f(z)$. Show that the necessary condition for analyticity of a function $f(z) = u + iv$ at $z = x + iy$ is to satisfy Cauchy Rie-mann equation 8

OR

Show that $v = 2xy - \frac{y}{x^2 + y^2}$ is a harmonic function find harmonic conjugate u of v . 5.

- b) State and prove Cauchy integral formula and by using it evaluate the integral $\oint_c \frac{z^2}{z^2 - 1} dz$ where c is positively oriented circle $|z-1| = 1$ 7

OR

State the Laurent theorem. If $f(z) = \frac{1}{z^2 - 1}$. Expand the function at $z = 1$ as Laurent series.

- a) Define z -transform. State and prove second shifting theorem of z -transform then evaluate $z(na^n)$ and $z(n^2-n)$ 8

OR

If $F(z)$ and $G(z)$ are respectively the z -transform of $f(t)$ and $g(t)$ then prove that:

$$Z[f(t) \times g(t)] = F(z) \cdot G(z)$$

- b) Solve the difference equation by using z - transform $y_{n+2} - y_n = 2^n$ where $y_0 = 0; y_1 = 1$ 7

3.

- a) Solve one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0,t)=0$ $u(l,t)$ and the initial condition $u(x,0)=f(x)$. 8

OR

Obtain the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar co-ordinates.

- b) Solve one dimensional wave equation with initial deflection is $0.01 \sin 3x$ and initial velocity is zero and $L = \pi$, $c^2 = 1$ 7

4. a) Using simplex method 8

Maximize $z = 90x_1 + 50x_2$

Subject to $x_1 + 3x_2 \leq 18$

$$x_1 + x_2 \leq 10$$

$$3x_1 + x_2 \leq 24$$

- b) Using the method of separation of variables find the solution $u(x, y)$ of the partial differential equation $u_x + u_y = (x + y)u$ 7

5.

- a) Show that $\int_0^\infty \left[\frac{\cos xw + w \sin xw}{1 + w^2} \right] dw = \begin{cases} 0 & \text{if } x < 0 \\ \pi / 2 & \text{if } x = 0 \\ \pi e^x & \text{if } x > 0 \end{cases}$ 8

- b) Find Fourier sine and cosine transform of $f(x) = 2e^{-5x} + 5e^{-2x}$ 7

OR

Let $f(x)$ be continuous on the x -axis and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and $f'(x)$ is absolutely integrable on x -axis then prove that

$$F[fx] = iwF[f(x)]$$

$$F[f'(x)] = -w^2 F[f(x)]$$

6.

- a) Find the inverse z -transform of $\frac{z^2 + z}{(z-1)(z^2+1)}$ 8

- b) Define singularity, zeros and poles of a function, Evaluate $\oint_c f(z) dz$ 7

where $f(z) = \frac{e^{2z}}{(z+1)^3}$ where c is the ellipse $4x^2 + 9y^2 = 16$.

7. Write short notes on:

- a) Find a tangent vector and the corresponding unit tangent vector 2.5
 $\vec{r}(t) = t\vec{i} + t^3\vec{j}$ at $P(1,1,0)$
- b) Represent the curve $x^2 + y^2 = 16$, $z = 3 \tan^{-1}(y/x)$ parametrically. 2.5
- c) Write the equation of ellipsoid and draw its rough sketch. 2.5
- d) Show that fourier cosine transform satisfied linearity property. 2.5

Level: Bachelor

Semester – Fall

Programme: BE

Course: Engineering Mathematics IV

Year : 2013

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Show that the function $u(x, y) = 3x^2y + x^2 - y^3$ is a harmonic function. Find its harmonic conjugate.

y^2 is a harmonic 8

b) State and prove Cauchy–integral formula and

pence evaluate 7

$\oint \frac{4 - \sin z}{z^2 - 2z} dz$, where c is the square with vertices

± 1 and $\pm i$

Or

Evaluate $\oint_C \frac{\cot z}{\left(z - \frac{\pi}{2}\right)^2} dz$, where C is the ellipse $4x^2 +$

$\partial y^2 = 36$. 7

a) State and prove Cauchy's Residue theorem.

b) Define Z-transform. State and prove first shifting theorem. And, evaluate Z-transform of $a^n \cos nt$ and

g theorem of Z- 8

$\sin nt$.

a) Find the inverse Z-transform of the function $\frac{z+2}{z^2-5z+6}$.

7

b) Solve the difference equation

$y_{n+2} - 8y_{n+1} + 16y_n = 4^n$, Where $y_0 = 0$ and $y_1 = 1$

8

a) Write one- dimensional wave equation and solve it, completely.

8

b) Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, having zero temperature in the end points and initial temperature $f(x)$.

7

a) What is Helmholtz's equation on $F(x, y)$ and solve it subject to $F(0, y)$

8

$$= 0 = F(a, y) = F(x, 0) = F(x, b).$$

- b) Define Fourier sine and cosine integrals. Show that 7

$$\int_0^{\infty} \frac{\cos wx + w \sin wx}{1+w^2} dx = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

Or

$$\text{Show that: } \int_0^{\infty} \left(\frac{1 - \cos \pi w}{w} \right) \sin wx dw = \begin{cases} \pi/2 & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

6. a) Using Simplex method, maximize the function $z = 4x_1 + x_2 + 2x_3$ 8
Subject to:

$$x_1 + x_2 + x_3 \leq 1$$

$$x_1 + x_2 - x_3 \leq 0$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- b) Define convolution of the two functions. State and prove convolution theorem on Fourier transform 7

7. Write Short notes on **any two**: 2×5

- Write the equation of hyperboloid of two sheet and then sketch
- Find the parametric representation of the surface $y^2 + (z-3)^2 = 9, x=2$.
- Show that Fourier sine transform is linear operator.
- Express $f(z) = \sin z$ in the form $u+iv$

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2013

Programme: BE

Full Marks: 100

Course: Engineering Mathematics IV

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define harmonic function. Show that $u(x, y) = \sin x \cosh y$ is harmonic and find corresponding analytic function. 8

- b) State and prove Cauchy Integral formula and use it to calculate:

$$\oint_c \frac{\cosh 3z}{5z} dz \quad \text{where } c: |z| = 1, \text{ counterclockwise.} \quad 7$$

- a) Define zeros and pole of a function. State Cauchy residue theorem. Evaluate: 8

$$\oint_c \left(\frac{z^2 \sin z}{4z^2 - 1} \right) dz \quad \text{where } c: |z| = 2, \text{ counter-clockwise.}$$

- b) Using the method of separation of variable solve the partial differential equation $u_{xx} + u_{yy} = 0$ 7

- a) Derive the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ 7

- b) Find the deflection function $u(x, t)$ of the vibrating string of length $L = \pi$ where $c^2 = 1$, the initial velocity is zero and the initial deflection is 8

$$\begin{cases} 0.01x & \text{if } 0 < x < \frac{\pi}{2} \\ 0.01(\pi - x) & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$$

4.

$$a) \text{ Show that } \int_0^{\infty} \frac{\cos ax \sin \omega}{\omega} d\omega = \begin{cases} \frac{\pi}{2} & \text{if } 0 \leq x < 1 \\ \frac{\pi}{4} & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

b) Find the Fourier transform of the function

$$f(x) = \begin{cases} |x| & \text{for } -1 < x < 1 \\ 0 & \text{for otherwise} \end{cases}$$

OR

Find the Fourier Cosine transform of the function $f(x) = e^{-kx}$ ($k > 0$)5. a) Define Z-transform of a function $f(t)$ and by using the definition find the Z-transform of

i. $(-1)^n$

ii. n

b) Solve the difference equation using z-transform

$$y_{n+2} - 3y_{n+1} + 2y_n = 4^n, y_0 = 0, y_1 = 1$$

6.

a) Find the inverse Z-transform of the function $\frac{2z^2 - 5z}{(z-2)(z-3)}$ b) Find $u(x, y, t)$ for the rectangular membrane with sides a and b with $c=1$, if the initial velocity is zero and the initial deflection is

$$\sin \frac{2\pi x}{a} \sin \frac{3\pi y}{b}$$

7. Solve the followings:

a) Find the Z-transform of discrete unit time impulse $\delta(n)$

b) Write down the equation of the ellipsoid and then sketch

c) Represent the curve $y^2 + (z-3)^2 = 9, x=0$ parametricallyd) Find the Imaginary part of z^2 .

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Mathematics IV

Semester: Fall

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define analyticity of a complex valued function $f(z)$. Show that the function $u = \frac{x}{x^2+y^2}$ is harmonic. Find harmonic conjugate of u such that $f(z) = u + iv$ is analytic. 8

- b) State and Prove Cauchy Residue theorem. Evaluate $\oint_C \frac{dz}{z^8(z+4)}$, where C is $|z+2| = 3$ in anticlockwise direction. 7

- a) Define conformal mapping. If $u = 2x^2+y^2$ and $v = \frac{y^2}{x}$ show that the curves $u = \text{constant}$ and $v = \text{constant}$ cut orthogonally at all intersections but the transformation $w = u + iv$ is not conformal. 7

OR

State and prove Cauchy–integral formula and hence evaluate

$$\oint_C \frac{2x^2 + 4z}{z-2} dz; \quad c: |z|=1$$

- b) Define Z transform. State and Prove first shifting theorem on Z transform. Using it find Z transform of $\cos at$ and $\sin at$. Also evaluate $Z(a^n \cos bt)$ and $Z(a^n \sin bt)$. 8

- a) Solve the difference equation: $y_{n+2} - 3y_{n+1} + 2y_n = 0$, where $y_0 = 0$ and $y_1 = 1$. 8

- b) Find the temperature in a laterally insulated bar of length L whose ends are kept at a zero temperature, assuming that the initial 7

temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L-x & \text{if } \frac{L}{2} < x < L \end{cases}$

4. a) Write one-dimensional wave equation and solve it. 8
 b) Using the method of separation of variable solve the partial differential equation $y^2 u_x - x^2 u_y = 0$. 7
5. a) Express Laplacian in polar co-ordinate system from Cartesian co-ordinate system. 8

OR

Find $u(x, y, t)$ for the rectangular membrane with sides a and b with $c = 1$, if the initial velocity is zero and initial deflection is

$$\sin \frac{2\pi x}{a} \sin \frac{3\pi y}{b}.$$

- b) Define Fourier sine and cosine integrals. Show that 7

$$\int_0^{\infty} \frac{\cos wx + w \sin wx}{1 + w^2} dx = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

OR

$$\text{Show that: } \int_0^{\infty} \left(\frac{1 - \cos \pi w}{w} \right) \sin wx dw = \begin{cases} \pi/2 & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}.$$

6. a) Find the Fourier transform of $f(x) = xe^{-x^2}$. 8
 b) State and prove initial and final value theorem on Z transform. 7
7. Answer the followings: 4×2.5
- a) Sketch the paraboloid $z = x^2 + y^2$
 b) Find the parametric representation of the surface $y^2 + (z-3)^2 = 9$, $x=2$
 c) Find the unit tangent vector of
 $\vec{r}(t) = \cos t \vec{i} + 2 \sin t \vec{j}$ at $\left(\frac{1}{2}, \sqrt{3}, 0 \right)$
 d) Show that $\oint_C \frac{dz}{z} = 2\pi i$, where C is the circle $|z| = 1$ in anticlockwise direction.

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Engineering Mathematics IV

Semester: Spring

Year : 2014
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analytic function $f(z)$. State Cauchy Riemann equation and hence show that it is the necessary condition for the function to be analytic. 7

- b) State and prove Cauchy's integral formula. Hence using it integrate 8

$$\oint_c \frac{z^2}{(z^4 - 1)} dz \text{ where } c \text{ is the circle } |z+i|=1 \text{ in counterclockwise.}$$

OR

Evaluate $\oint \frac{z^3 + \sin z}{c(z-i)^3} dz$, where 'c' is the boundary of the square with vertices $\pm 2, \pm 2i$.

2. a) Expand the function $f(z) = \frac{z+3}{z(z^2 - z - 2)}$ in the region given by 8

- i. $|z| < 1$,
- ii. $1 < |z| < 2$,
- iii. $|z| > 2$.

- b) Find the deflection $u(x, t)$ of the vibrating string of length $L = \pi$, $c^2 = 1$ and its initial velocity is zero and initial deflection is given by 7

$$f(x) = \begin{cases} 0.1x, & \text{for } 0 < x < \frac{\pi}{2} \\ 0.01(\pi - x), & \text{for } \frac{\pi}{2} < x < \pi \end{cases}$$

3. a) Find the solution of one dimensional wave equation by D' Alembert's method.

b) Express the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar co-ordinates.

4. a) State and prove first shifting theorem for Z-transform. Use it to find $Z(\cosh at \sin bt)$.

b) Find the inverse Z-transform of $\frac{2z}{(z-1)(z^2+1)}$

5. a) Show that $Z(y_{n+k}) = z^k \left[\bar{y} - y_0 - \frac{y_1}{z} - \dots - \frac{y_{k-1}}{z^{k-1}} \right]$ where $\bar{y} = Z(y_n)$.

Using it solve $y_{k+1} + y_k = 1$ if $y_0 = 0$.

b) Solve $u_{xx} + u_{yy} = 0$ by using separation method.

6. a) Define convolution of two functions. State and prove convolution theorem on Fourier transform.

OR

Define Fourier transform and evaluate Fourier transform of

$$f(x) = e^{-\frac{x^2}{2}}$$

b) Derive Fourier integral of $f(x)$ from Fourier series. Show that.

$$\int_0^\infty \left[\frac{\cos xw + w \sin xw}{1+w^2} \right] dw = \begin{cases} 0 & \text{if } x < 0 \\ \pi/2 & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

7. Write short notes on:

a) If $\mathbf{r}(t) = (a + 2\cos 2t, b - 2\sin 2t, 0)$ be the position vector of any curve, find its equation in Cartesian form.

b) Verify that $u = x^2 + t^2$ is the solution of one dimensional wave equation.

c) Define the types of singularity of a complex function with examples.

d) Find $Z(1)$ and $Z(-1)^n$

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define analytic function $f(z)$. State Cauchy Riemann equation and hence show that it is the necessary condition for the function to be analytics. 7
- b) State and prove Cauchy's integral formula. Evaluate where c is the ellipse $4x^2 + 9y^2 = 36$. 8
- $$\oint_c \frac{\cot z}{\left(z - \frac{\pi}{2}\right)^2} dz$$

OR

Evaluate $\oint_c \frac{z^3 + \sin z}{(z - i)^3} dz$, where 'c' is the boundary of the square with vertices $\pm 2, \pm 2i$.

- a) State Laurent's theorem. Find the Laurent's series for $f(z) = \frac{1}{(z^2 - z^3)}$ in the region $0 < |z| < 1$. 7
- b) Define singularity, zeros and poles of a function. Evaluate $\oint_c f(z) dz$ where $f(z) = \frac{e^{2z}}{(z + 1)^3}$ where c is the ellipse $4x^2 + 9y^2 = 16$. 8
- a) State and prove convolution theorem on Z transform. 7
- b) Solve the difference equation: $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$, where $y_0 = 0$ and $y_1 = 0$. 8
- a) Find the Fourier integral of the function; 7

$$f(x) = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1 \\ \pi/4 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

8

- b) Find Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$, and then

$$\text{show that } \int_0^{\infty} \left(\frac{\cos kx}{1+x^2} \right) dx = \frac{\pi}{2} e^{-k}$$

5. a) Derive one dimensional wave equation of a string of length L which is fixed in two end points with required assumptions.

7

OR

Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, with initial temperature $f(x)$ and boundary conditions is $u(0,t)=0=u(L,t)$.

- b) Derive two dimensional heat equations with necessary assumptions.

8

6. a) Find $Z^{-1} \left[\frac{2z^2 + 3z}{(z+2)(z-4)} \right]$

7

- b) A homogeneous rod of conducting material of length 100 cm has its end kept at zero temperature and temperature initially is

$$f(x) = \begin{cases} x, & 0 \leq x \leq 50 \\ 100 - x, & 50 \leq x \leq 100. \end{cases}$$

8

7. Write short notes on: (Any two)

- a) Find z -transform of na^{n-1}

2.5×
4=10

- b) Evaluate $\oint_C \frac{z^3}{2z-i} dz$ where $|z|=1$.

- c) Solve the partial differential equation: $u_x + u_y = 0$, by separation of variables method.

- d) Write equation of ellipsoid. Sketch it with center and axes of symmetry.

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2015

Programme: BE

Full Marks: 100

Course: Engineering Mathematics IV

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Show that the necessary condition for analyticity of $f(z) = u+iv$, is $u_x = v_y$ and $u_y = -v_x$. 7
- b) Define Laplace equation. Test $u = \cos x \cosh y$ is harmonic or not. If yes, find the harmonic function and the corresponding analytic function $f(z)$. 8
2. a) State and Prove Cauchy Residue theorem. Evaluate $\oint_C \frac{e^{5z}}{(z+i)^4} dz$, where C is a circle $|z| = 3$ along anticlockwise direction. 7
- b) Determine the region of $w = e^{\frac{i\pi}{4}}$ in the w -plane corresponding to the triangular region bounded by the lines $x=0$, $y=0$, and $x+y=1$ in the z -plane. 8

Or

Integrate: $\oint_C \frac{dz}{z^2 + 4}$; $C : 4x^2 + (y-2)^2 = 4$

3. a) Find the Z-transform of $f(t) = a^n$ and hence find $Z \left\{ \sin\left(\frac{n\pi}{2}\right) \right\}$ and $Z \left\{ \cos\left(\frac{n\pi}{2}\right) \right\}$. 7
- b) Find the inverse of z-transform of $\frac{3z^3 + 2z}{(z-3)^2(z-2)}$. 8
4. a) Solve the difference equation: $y_{n+2} + 6y_{n+1} + 9y_n = 4^n$, where $y_0 = 0$ and $y_1 = 0$. 7

- b) Find Fourier sine and cosine integral representation of the function
 $f(x) = e^{-x} + e^{-2x}$, for $x > 0$.

Or

Find Fourier transform of $f(x) = \begin{cases} 1 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$.

5. a) Define partial differential equation with suitable example. By separating the variables solve $u_{xx} + u_{yy} = 0$
- b) Define partial differential equation with suitable example. By separating the variables solve $u_{xx} + u_{yy} = 0$

Or

Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, with initial temperature $f(x)$ and boundary conditions is $u(0,t)=0=u(L,t)$.

6. a) Express the Laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar co-ordinates.

Or

Solve one dimerisional wave equation Completely.

- b) Define Fourier integral. Choosing a suitable function, show that

$$\int_0^{\infty} \frac{\sin \pi \omega}{\omega} \sin \omega x d\omega = \begin{cases} \frac{\pi \sin \pi x}{2} & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

Or

Find the Fourier cosine transform of e^{-x} .

7. Write short notes on: (Any two)

- a) Solve the partial differential equation $u_x = 2xy$.
- b) Write equation of an ellipsoid. Sketch it with centre and axis of symmetry.

- c) Verify that $u = x^2 + t^2$ is the solution of one dimensional wave equation

- d) Derive Z inverse of $X(z) = \frac{z}{(z+1)(z-3)}$.

Level: Bachelor
 Programme: BE
 Course: Engineering Mathematics IV

Semester: Fall

Year : 2016
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Prove that if the function $f(z)$ is analytic then show that $U_x = V_y$ and $U_y = -V_x$ 8
- b) Integrate the followings along the unit circle counterclockwise 7
- i. $\oint \frac{z^6}{(2z-1)^6}$
- ii. $\oint \frac{z+1}{z^3-2z^2}$
- a) Find the singular points and residues of the function 8
- $$f(z) = \frac{z+2}{(z-2)(z^2+1)^2}$$
- b) State Laurent's theorem. Find the Laurent's series for 7
- $$f(z) = \frac{1}{(z-z^3)}$$
- in the region
- $0 < |z+1| < 2$
- .
- a) Find the Z- transform of the function $f(t) = e^{-iat}$ and hence deduce the value of $Z(\cos at)$ and $Z(\sin at)$. 7
- b) Using Z-transform solve the difference equation 8
- $$y_{n+2} + 6y_{n+1} + 9y_n = 2^n \text{ when } y_0 = y_1 = 0.$$
- a) Define Z - transform. State and prove Second shifting theorem of Z- transform. Evaluate $Z(t^2 e^{-bt})$ 7

OR

Find $Z^{-1} \frac{z^2 + 1}{z^2 - 2z + 2}$.

- b) Choosing a suitable function show that $\int_0^\infty \left[\frac{\cos \omega x + \omega \sin \omega x}{1 + \omega^2} \right] d\omega =$ 8

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2016

Programme: BE

Full Marks: 100

Course: Engineering Mathematics IV

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.
 - a) Define analyticity of the complex valued function $f(z)$. If $f(z) = z + \frac{1}{z}$, check analyticity of $f(z)$ by using Cauchy Riemann equation. 8
 - b) State and prove Cauchy integral formula. Integrate $\oint_C \frac{1}{z^2 + 4} dz$, 7
 $C: 4x^2 + (y - 2)^2 = 4$ counter clock wise.
2. a) Obtain the Taylor series and Laurent series of the function 7
 $f(z) = \frac{1}{(z+2)(z^2+1)}$ when $1 < |z| < 2$.

OR

Define conformal mapping. Name the types of conformal mappings. Translate the rectangular region ABCD in Z plane bounded by $x=1$, $x=3$, $y=0$ and $y=3$ under the transformation $w=z+(2+i)$. Illustrate with figure also.

- b) State Cauchy Residue Theorem and hence evaluate $\oint \frac{z-23}{z^2-4z-5} dz$ where 8
 $C: |z-2|=4$.
3. a) Obtain the Fourier integral formula from the Fourier series assuming the required conditions. 7

OR

Show that: $\int_0^\infty \left(\frac{1 - \cos \pi w}{w} \right) \sin wx dw = \begin{cases} \pi/2 & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$

- b) Find the Fourier transform of the function 8

$$f(x) = \begin{cases} 1 - x^2 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$$

4. a) Find the solution of one dimensional wave equation by using D'Alembert's method. 8
- b) Find the temperature distribution in a laterally insulated thin copper bar ($c^2 = 1.158 \text{ cm}^2/\text{sec}$), 100cm long and of constant thickness whose end points at $x = 0$ and $x = 100$ are kept at 0°C and initial temperature is $f(x) = \sin^3(0.01)\pi x$ 7
5. a) A string of length 20cm is fastened at both ends is displaced from its position of equilibrium by imparting to its points an initial velocity $g(x) = \begin{cases} x & \text{if } 0 \leq x \leq 10 \\ 20 - x & \text{if } 10 \leq x \leq 20 \end{cases}$ 7
- Find the deflection $U(x, t)$
- b) Derive two dimensional heat equation and solve completely. 8
5. a) State and prove first and second shifting theorems in Z-transform. Find the value of $Z(a^n \cos bt)$ and $Z(a^n \sin bt)$. 8
- b) Using Z-transform, solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ when $y_0 = y_1 = 0$. 7
7. Attempt all: 2.5×4
- a) If $z = u + iv$ is an analytic function then prove that u and v both satisfy Laplace equation
- b) Represent the curve $y^2 - (z-3)^2 = 9$, $x = 0$ parametrically
- c) Evaluate $\oint \frac{z^3 \sin z}{3z-1} dz$ along a unit circle
- d) State and prove the linear property on Z-transform

$$\begin{cases} 0 & \text{if } x < 0 \\ \pi/2 & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

5. a) Find the Fourier cosine and sine transform of $f(x) = e^{-ax}$, $a > 0$.

OR

Verify the convolution theorem for the functions $f(x) = e^{-x^2}$ and

$$g(x) = e^{-x^2}.$$

- b) Find $u(x, t)$ of the string of length $l = \pi$ when $c^2 = 1$, the initial velocity is zero and the initial deflection is $0.1(\pi - x)$.
6. a) What is Helmholtz's equation on $F(x, y)$ and solve it subject to $F(0, y) = 0 = F(a, y) = F(x, 0) = F(x, b)$.

OR

Find the deflection $u(x, y, t)$ of the square membrane with $a = b = 1$ and $c = 1$, if the initial velocity is zero and the initial deflection is $(0.1) \sin 3\pi x \sin 4\pi y$.

- b) Derive one dimensional heat equation with required assumptions.
7. Attempt all

- a) Solve by using separation of variables $u_x - u_y = 0$
- b) Examine whether $\bar{z}$ is analytic or not?
- c) Find the unit tangent vector to the curve

$$\vec{r}(t) = 2 \cos t \vec{i} + \sin t \vec{j} \text{ at } (\sqrt{2}, \sqrt{2}, 0).$$

- d) Sketch the paraboloid $z = x^2 + y^2$.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analytic function. Show that the function $u(x, y) = 3x^2y + x^2 - y^3 - y^2$ is a harmonic function. Find a function $v(x, y)$ such that $u + iv$ is an analytic function. 8

- b) Define Pole and Zeroes of a function. State Cauchy's residue theorem 7

and evaluate $\oint_C \frac{e^z}{\cos z} dz$ where $C : |z| = 3$.

2. a) Find the expansion of $\frac{7z - 2}{(z + 1)z(z - 2)}$ in the region given by 8

i) $0 < |z + 1| < 1$. ii) $1 < |z + 1| < 3$.

- b) Given the bilinear transformation $w = \frac{3 - z}{2z + 1}$, find the mapping of the circle $|z| = 1$ in the w -plane 7

3. a) Show that $\int_0^\infty \frac{\cos wx + w \sin wx}{1 + w^2} dx = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$ 8

- b) Find the Fourier sine and cosine transform of the function $f(x) = 2e^{-5x} + 5e^{-2x}$ 7

4. a) Derive and find the solution of one dimensional wave equation. 8

- b) Find the temperature in a laterally insulated bar of length L whose ends are kept at a zero temperature, assuming that the initial 7

temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \end{cases}$

Level: Bachelor

Programme: BE

Course: Engineering Mathematics IV

Semester spring

Year : 2018

Full Marks: 100

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analytic function. Show that the function $u(x, y) = 3x^2y + x^2 - y^3 - y^2$ is a harmonic function. Find a function $v(x, y)$ such that $u + iv$ is an analytic function. 8

- b) Define Pole and Zeroes of a function. State Cauchy's residue theorem and evaluate $\oint_C \frac{e^z}{\cos z} dz$ where $C : |z| = 3$. 7

2. a) Find the expansion of $\frac{7z - 2}{(z + 1)z(z - 2)}$ in the region given by 8
i) $0 < |z + 1| < 1$. ii) $1 < |z + 1| < 3$.

- b) Given the bilinear transformation $w = \frac{3 - z}{2z + 1}$, find the mapping of the circle $|z| = 1$ in the w -plane 7

3. a) Show that $\int_0^\infty \frac{\cos wx + w \sin wx}{1 + w^2} dx = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$ 8

- b) Find the Fourier sine and cosine transform of the function $f(x) = 2e^{-5x} + 5e^{-2x}$ 7

4. a) Derive and find the solution of one dimensional wave equation. 8
b) Find the temperature in a laterally insulated bar of length L whose ends are kept at a zero temperature, assuming that the initial 7

temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$

5. a) Solve $U_{xx} + U_{yy} = 0$ by the method of separation of variables 8
 b) What is Helmholtz equation? Find its solution. 7
6. a) State and prove Initial and Final value theorems in Z-transform. Find the value of $Z(a^n \cos bt)$ and $Z(a^n \sin bt)$ 8
 b) Solve $y_{n+2} - 3y_{n+1} + 2y_n = 0$, $y_0 = 0, y_1 = 1$ 7
7. Answer all of the following questions. 4×2.5
- a) Express the parametric equation of the hyperbola $x^2 - y^2 = 1, z = 0$.
 b) Check the analyticity of the function $f(z) = \text{Arg } z$
 c) Find the z-transform of $f(n) = na^n$.
 d) Find the residue of $f(z) = \frac{1}{z^2 - 1}$ at $z = 1$.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define harmonic function. Show that the function $u = 3x^2y + x^2 - y^3 - y^4$ is a harmonic function. Find the analytic function for which the given function is a real part. 8

- b) Evaluate $\oint_C \left(\frac{\cos(\pi z^2)}{z^2 - 3z + 2} \right) dz$ where $C: |z| = 3$. 7

- a) Let the rectangular region R in the z -plane be bounded by lines $x=0$, $y=0$, $x=2$, $y=3$. Find the region R' of the w -plane into which R is mapped under the transformation $W = \sqrt{2} e^{\frac{i\pi}{4} z}$. 7

- b) Find the Taylor's and Laurent's series of the function 8
- $$f(z) = \frac{z^2 - 1}{(z + 2)(z + 3)}$$

OR

State Cauchy Residue Theorem. By applying Cauchy Residue

Theorem, evaluate $\oint_C \left(\frac{4 - 3z}{z(z - 1)(z - 2)} \right) dz$ where $C: |z| = \frac{3}{2}$.

- a) State & prove first shifting theorem on Z-transform. Find the Z-transform of $e^{-at}t$. 8

- b) Solve the differential equation $y_{k+2} + 2y_{k+1} + y_k = k$ where $y_0 = 0$, $y_1 = 0$ using Z-transform. 7

- a) Show that $\int_0^\infty \frac{\sin \pi w \sin xw}{1 - w^2} dw = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$ 7

- b) Find the Fourier cosine transform of $f(x) = e^{-x}$ ($x > 0$) and hence by using Parseval's identity, show that that $\int_0^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$. 8
5. a) Define partial differential equation with suitable example. By separating the variables solve $u_{xx} + u_{yy} = 0$ 7
- b) Find $u(x, t)$ from one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, with boundary condition. $u(0, t) = 0 = u(L, t)$, initial deflection $f(x)$ and initial velocity $\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x)$. 8
6. a) Find the temperature in a laterally insulated bar of length L whose ends are kept at a zero temperature, assuming that the initial temperature is $f(x) = \begin{cases} x & \text{if } 0 < x < \frac{L}{2} \\ L - x & \text{if } \frac{L}{2} < x < L \end{cases}$ 7
- b) Express the laplacian $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ in polar co-ordinates. 8
7. Attempt all questions. 4×2.5
- a) If $u = y^3 - 3x^2y$ show that u is harmonic.
- b) Find z -transform of $z(a^n)$
- c) If $\vec{r} = (3\cos t, 4\sin t, t)$ be the position vector of the curve. Find its curve.
- d) Solve the partial differential equation $u_{yy} = u$.

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2019

Programme: BE

Full Marks: 100

Course: Engineering Mathematics IV

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by analyticity of function $f(z)$. State Cauchy Remann equation and hence show that it is the necessary condition for the functions to be analytic. 8
 b) State Cauchy Integral formula. Evaluate $\int \frac{1}{z^2+4} dz$, where integration is along the ellipse $4x^2 + (y-2)^2 = 4$ 7
2. a) Find the image of infinite strip $1/4 < y < 1/2$ under the transformation $w = 1/z$. 8
 b) Define Singularities of a function $f(z)$. Find the residues of $f(z) = \frac{z+2}{(z+1)(z^2+1)^2}$. 7
3. a) Find the inverse z- transform of $f(z) = \frac{2z}{(z-1)(z^2+1)}$ 7
 b) Using Z-transform solve the difference equation $y_{k+2} + 2y_{k+1} + y_k = k$, where $y_0 = 0, y_1 = 0$ 8
4. a) Derive Fourier integral of $f(x)$ from Fourier series. Show that: 7

$$\int_0^\infty \frac{\cos xw}{1+w^2} dw = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

 b) Find Fourier cosine transform of $f(x) = e^{-mx}$ for $m > 0$. Then prove 8
 that $\int_0^\infty \frac{\cos kx}{1+x^2} dx = \frac{\pi}{2} e^{-k}$
5. a) Using the method of separation of variable solve P.D.E.: $y^2 u_x - x^2 u_y = 0$ 7

- b) Find $u(x, t)$ from one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, with 8
 boundary condition, $u(0, t) = 0 = u(L, t)$, initial deflection $f(x)$ and
 initial velocity $\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x)$.

6. a) A homogeneous rod of conducting material of length 100 cm has its 8
 ends kept at zero temperature and the temperature initially is

$$f(x) = \begin{cases} x, & 0 \leq x \leq 50 \\ 100 - x, & 50 \leq x \leq 100 \end{cases}$$

Find the temperature $u(x, t)$

- b) Find two-dimensional Laplace equation in polar co-ordinates. 7

7. Attempt all the questions. 4×2=

- a) Verify that: $U = x^2 + t^2$ is the solution of one dimensional wave equation 5

- b) Find the Z- transform of na^{n-1}

- c) Check analyticity of $f(z) = z^3$

- d) Find the unit tangent vector to the curve

$$\vec{r}(t) = 2 \cos t \vec{i} + \sin t \vec{j} \text{ at } (\sqrt{2}, \sqrt{2}, 0).$$

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2020

Programme: BE

Full Marks: 100

Course: Engineering Mathematic IV

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Check $u = \sin x \cosh y$ is harmonic or not? If yes, find corresponding harmonic conjugate v of u 8
 b) Evaluate $\oint_C \frac{\cot z}{\left(z - \frac{\pi}{2}\right)^2} dz$, where C is the ellipse $4x^2 + 9y^2 = 36$. 7
2. a) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent series valid for 7
 (i) $1 < |z| < 3$ (ii) $|z| > 3$ (iii) $|z| < 1$ (iv) $0 < |z+1| < 2$
 b) State and prove Cauchy residue theorem. Using it evaluate 8
 $\int_C \left(\frac{z^2 \sin z}{4z^2 - 1} \right) dz$ where C is the circle $|z| = 2$. 7
- a) Find the Z trans form of (i) $r^n \cos n\theta$ (ii) $\frac{1}{n+2}$. 7
- b) Solve the differential equation $y_{k+2} + 2y_{k+1} + y_k = k$ where $y_0 = 0$, $y_1 = 0$ using Z-trans form. 8
4. a) Find the solution of the differential equation, $y^2 u_x - x^2 u_y = 0$, by using separating of variables. 7
 b) Find the solution of one dimensional equation with boundary 8
 $u(0, t) = 0 = u(l, t)$ and initial condition $u(x, 0) = \left(\frac{100x}{l} \right)$.

5. a) Derive one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ with necessary assumptions. 7

- b) What is Helmholtz's equation on $F(x, y)$ and solve it subjected to $F(0, y) = 0 = F(x, 0) = F(x, b)$. 8

"OR"

Find the deflection $u(x, y, t)$ of the square membrane with $a = b = 1$ and $c = 1$, if the initial velocity is zero and the initial deflection is $(0, 1)$.

$$\sin 3\pi x \sin 4\pi y.$$

6. a) Find the Fourier transform of $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ 7

- b) Define Fourier cosine integral. Hence, show that 8

$$\int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1 \\ \pi/4 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

7. Write short notes on: 10

- a) Write down Laplacian in cylindrical coordinate systems.

b) Prove $Z(a^n) = \frac{z}{z-a}$

- c) Show that the transformation $w = e^z$ is conformal

- d) Show that the Fourier cosine transform satisfied linearity property.